

To address this, we developed a GNN model that bypasses the limitations of brute-force calculations by providing rapid physics predictions. Although trained on a relatively small portion of the compositional space of the ternary Nb-Mo-Ta family of body-centered cubic (BCC) alloys, the GNN model accurately predicts fundamental quantities, including the Peierls barrier and solute/screw interaction energy. Several examples are provided to demonstrate the proficiency of this multi-agent approach in not only exploring the design space but also tackling complex alloy design challenges, such as predicting macroscopic yield stress.

The plan of the paper is as follows. We start by providing details about our GNN model to predict atomic-level material properties in Section 2. We then discuss our proposed LLM-driven multi-agent system for automated materials design, guided by the predictions from our GNN model. We outline the main components of the multi-agent approach and provide several examples to illustrate the proficiency of our model in solving complex alloy design problems. We then present the key findings in Section 3 and discuss the implications of our multi-agent system for future research in materials discovery.

2 Results and Discussion

The main tool implemented in our multi-agent system is the GNN model that accurately predicts the fundamental material properties in multi-component alloys, i.e. Peierls barrier and potential energy change due to the screw dislocation movement. We first delve into the basics of development of the GNN model and then discuss our proposed LLM-based multi-agent system to design alloys soliciting physics data from the GNN model.

2.1 GNN model

The workflow of the GNN approach for predicting the Peierls barrier and energy changes due to dislocation movement is schematically presented in Figure 1. Initially, the dislocated structures for 200 samples per composition within the training compositional space are generated, as shown in Figure S1 in the SI. After minimizing these structures, the potential energy change is calculated by subtracting the final potential energy from the initial one. Additionally, the Peierls barrier for each sample is extracted from the minimum energy path connecting the initial and final states, as derived from NEB simulations. The potential energy changes and the Peierls barriers serve as the ground truth or labels used to train the deep learning (DL) model. In this model, each random configuration is represented as a graph, where nodes correspond to atoms and edges to chemical bonds. Node features consist of chemical (solute types) and configurational features (screw dislocation displacement). The solute types at each site are one-hot encoded, with vectors $[1, 0, 0]$, $[0, 1, 0]$, and $[0, 0, 1]$ representing *Nb*, *Mo*, and *Ta*, respectively. Additionally, the z -component of the screw dislocation displacement, δ_z , is calculated for pure Mo and included as a node feature. Notably, only the chemical features vary across compositions, while the δ_z feature remains constant, thus eliminating the need for atomic relaxation in new random configurations during inference. Furthermore, edge features are constructed based on the solute types of the neighboring atoms. After constructing the graph input, the GNN model is trained to predict either the potential energy change or the Peierls barrier, which serves as the label for the entire graph. Once trained, the model can predict the potential energy change and Peierls barrier for new structures, drastically reducing the need for time-consuming atomistic simulations, particularly NEB calculations. This enables us to explore the full compositional space of NbMoTa ternary alloys to identify candidates with enhanced properties, such as a higher Peierls barrier.

Peierls barrier and potential energy change predictions are graph regression tasks. In this work, we adopt a GNN with the Principal Neighborhood Aggregation graph convolution operator (PNAConv),[53] which has outperformed many popular GNN models in the literature, such as GCN,[54] GAT,[55], and GIN [56] on benchmark graph regression tasks. The PNA model combines multiple aggregators, which affect how messages between nodes are passed, with a degree-scaler that generalizes the sum aggregator. Specifically, the architecture employs four different aggregators: *mean*, *maximum*, *minimum*, and *standard deviation*. The improved performance of the GNN is attributed to this combination, where the degree-based scalers amplify or attenuate signals in the network based on node degree.

More details about the construction of the dataset, the architecture of the GNN model, and the training procedure are provided in the Methods section. In the following sections, we evaluate the model’s performance in predicting material properties on the test dataset and assess its ability to generalize to unseen configurations.